

NAME: _____

PERIOD: _____

DATE: _____

Which Design Supports Cause?

AP Statistics · Topic 3.5 (CED 1.13) · B: 2026-10-02 · E: 2026-10-14

[STAR] TODAY'S PROBLEM

Suppose a researcher recruits 30 student volunteers to study whether an energy drink changes reaction time.

Design 1: The first 15 students to sign up receive the energy drink; the last 15 receive water. A tester who knows each drink measures reaction time.

Design 2: A coin flip assigns each volunteer to the energy drink or water. The water is flavored to match, both drinks are served in identical opaque cups, and neither the volunteer nor the tester knows which drink was assigned.

Two energy-drink designs

#	Assign	Control	Blind
1	Signup	Water	No
2	Coin	Matched	Both

FIRST TAKE — before the video (5 min)

Which design would make you more willing to attribute a reaction-time difference to the drink? Name one feature behind your choice.

[BOOK] PRINCIPLES OF EXPERIMENTAL DESIGN (keep on the page)

A well-designed experiment uses **comparison**, **random assignment**, **replication**, and control of potential confounding variables. A **control group** supplies a baseline. Random assignment tends to balance uncontrolled variables so a difference can be attributed to treatment. **Blinding** reduces changes caused by expectations or by how responses are measured. Selection still limits generalization.

Finish after the video:

DOK 1 (a) Identify the experimental units, explanatory variable and its levels, and response variable.

DOK 2 (b) For each design, state whether it has random assignment, a control group, and blinding. Explain one job performed by each feature.

DOK 3 (c) Which design could support the claim that the energy drink caused a reaction-time difference? Cite the specific features that settle it, explain what a difference under Design 1 could be due to instead, and decide whether either result could automatically generalize to all teenagers.

Frame: Design _____ supports cause because _____.

Frame: The result cannot automatically generalize to _____ because _____.

EXIT — turn this in

One thing the video changed about my first take: _____